UNIT-IV
FUEL TECHNOLOGY

Definition

A fuel is defined as naturally occurring or artificially manufactured combustible carbonaceous material which serves particularly as source of heat and light and also in few cases as a source of raw material.

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Classification of fuels Fuels are classified into a two types.

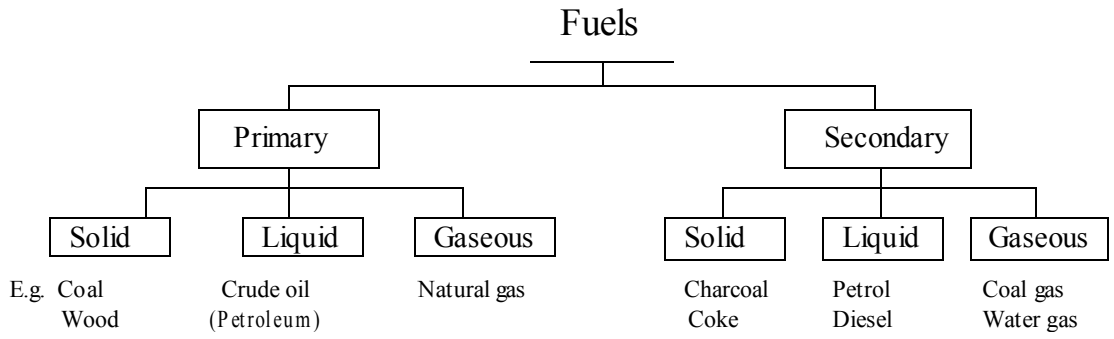
1) Based on their origin they are classified into
   a) Primary fuels
   b) Secondary fuels.

a) **Primary Fuels:** There are naturally occurring fuels which serves as source of energy without any chemical processing.
Ex: Wood, Coal, Crude oil, Natural gas, Peat, Lignite, Anthracite.

b) **Secondary Fuels: -** They are derived from primary fuels & serves as source of energy only after subjecting to chemical processing.
Ex: Charcoal, Coke, produsergas, Petrol, Diesel etc.,

Bases on their physical state fuel are classified into

a) Solid
b) Liquid
c) Gaseous fuels.



Liquid Fuels:

Liquid fuels are the important commercial and domestic fuels used in our daily life. Most of these fuels are obtained from the naturally occurring petroleum or crude oil called as primary fuel.

Advantages:

(a) They posses higher calorific value per unit mass than solid fuels.
(b) They burn without dust, ash, clinkers, etc.
(c) Their firing is easier and also fire can be extinguished easily by stopping liquid fuel supply.
(d) They are easy to transport through pipes.
(e) They can be stored indefinitely without any loss.
(f) They are clean in use and economic to handle.
(g) Loss of heat in chimney is very low due to greater cleanliness.
(h) They require less excess air for complete combustion.
(i) They require less furnace space for combustion.

Disadvantages:

(a) The cost of liquid fuel is relatively much higher as compared to solid fuel.
(b) Costly special storage tanks are required for storing liquid fuels.
(c) There is a greater risk of five hazards, particularly, in case of highly inflammable and volatile liquid fuels.
(d) They give bad odour.
(e) For efficient burning of liquid fuels, specially constructed burners and spraying apparatus are required.

Petroleum (crude oil)***:***

The crude oil or petroleum is also known as rock oil or mineral oil. It is dark colour liquid found well deep in the earth. It is mainly composed of hydrocarbons which may be solids, liquids or gases. Some optically active compounds having
S and N are also present. On the average the composition of

Petroleum is; C = 79.5 – 87.1%
H = 11.5 – 14.8%
S = 0.1 – 3.5%
N and O = 0.1 – 0.5%

Refining of petroleum:

Crude oil obtained from the mine is not fit to be marketed. It contains a lot of soluble and insoluble impurities, which must be removed. Previously the purification of crude oil is done by simple fractional distillation. Further treatment of the products is done by refining.

Definition:

"The process by which petroleum is made free of impurities, followed by the separation of petroleum into different fractions having different boiling points, and their further treatment to impart specific properties" can be called as Refining of petroleum.

Various stages during the refining process are explained below

1. Removal of solid impurities.
2. Removal of water.
3. Removal of harmful impurities.
4. Fractional distillation.

1. Removal of solid impurities:

The crude oil is a mixture of solid, liquid and gaseous substances. This is allowed to stand undisturbed for some time, till the heavy solid particles settle down and gases evaporate. The supernant liquid is then centrifuged where in the solids get removed.

2. Removal of water by Cottrell's process:

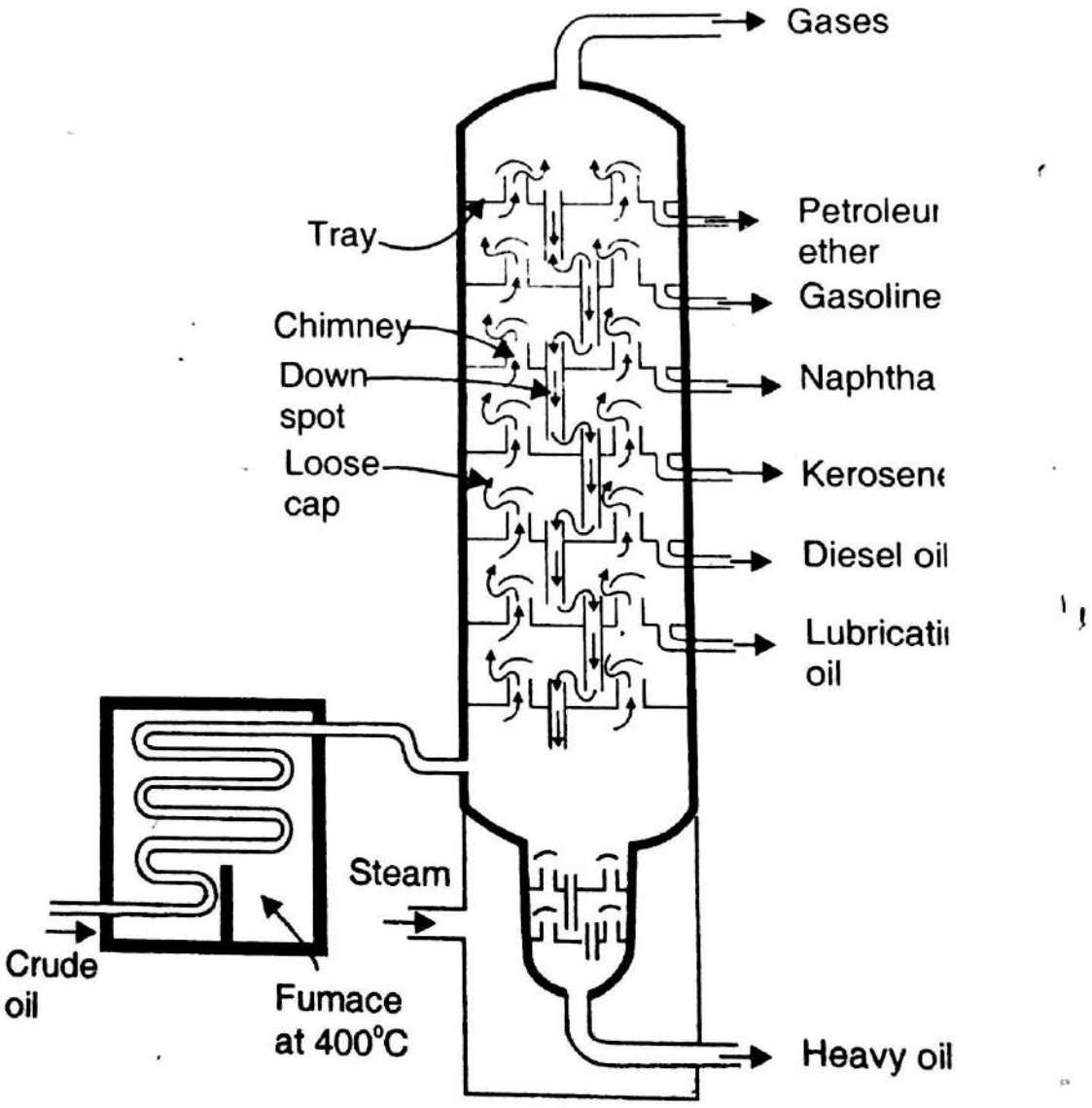
The crude oil obtained from the earth's crust is in the form of stable emulsion of oil and brine. This mixture, when passed between two highly charged electrodes, destroy the emulsion films and the colloidal water droplets coalesce into bigger drops and get separated out from the oil.

3. Removal of harmful impurities:

In order to remove sulphur compounds in the crude oil, it is treated with copper oxide. Then, the sulphur compounds get converted to insoluble copper

sulphide, which can be removed by filtration Substances like NaCl and $MgCl_2$, if present will corrode the refining equipment and result in scale formation. These can be removed by techniques like electric desalting and dehydration

Fractional distillation:



The crude oil is fractionally distilled in a fractionating still, which is a tower like structure having a bubble tower laid inside. The crude oil or the petroleum enters through the bottom of the still by an inlet. This is made to pass through a pre-heater or it can heated with steam inside the still to a temperature of about 400^0C. As we move to the top fractionating still the temperature goes on decreasing. A number of side pipes are fitted at different heights of the column or still, to collect the fractions. As petroleum enters the still, it gets vaporized and the

vapours move up the bubble tower. The bubble tower consists of number of horizontal trays, and each tray is provided with a number of small chimneys through which the vapours rise. The chimneys are covered with loose caps so that the vapours bubble through and get condensed, dropping back into the trays. So as the vapours go up, they become cooler and cooler and fractional condensation of the crude oil takes place at different levels of the bubble tower.

The vapours of the liquid having the highest boiling point are the first to condense and go out and those which have the lowest boiling points go last, along with the uncondensed gases. The constituents of each fraction and the temperature at which they are obtained is given in the table.

Name of fraction	***Boiling range***	***Composition***	***Uses***
1. Uncondensed gases	Below 30^0C	C_1 to C_4	As domestic or industrial fuel under the name L.P.G.
2. Petroleum ether	30 – 70^0C	C_5 to C_7	As a solvent
3. Gasoline or petrol(11,250 kcal/kg)	40 – 120^0C	C_5 to C_9	As a motor fuel in I.C. engine
4. Naphtha (11,000 kcal/kg)	120 – 180^0C	C_9 to C_{10}	As a solvent in dry cleaning
5. Kerosene oil (11,100 kcal/kg)	180 – 250^0C	C_{10} to C_{16}	As domestic fuel and jet engine fuel
6. Diesel oil	250 – 320^0C	C_{10} to C_{18}	Diesel engine fuel
7. Heavy oil This on refraction gives; (a) Lubricating oil (b) Petroleum jelly (Vaseline) (c) Grease (d) Paraffin wax	320 – 400^0C	C_{17} to C_{30}	 As lubricant In cosmetics and medicines. As lubricant In candles and wax paper
8. Residue may be either; (a) Asphalt Or	Above 400^0C	C_{30} and above	 Water proofing of roofs As a fuel

(b) Petroleum coke			

Synthetic petrol:

Petrol can be synthesised by any of the following methods.

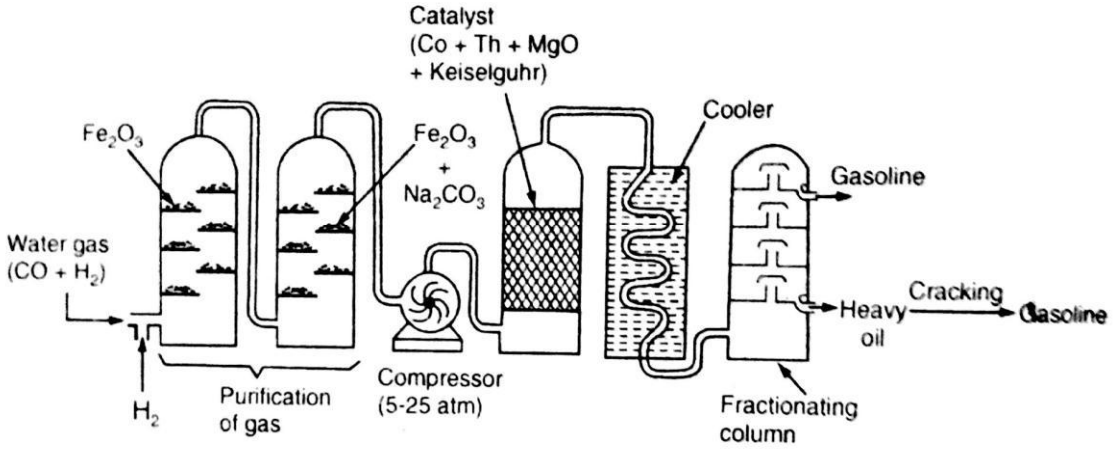
1. Fischer – Tropsch method.

2. Bergius process.

Fischer – Tropsch method:

Water gas (CO + H_2), produced by passing steam over heated coke, and is mixed with hydrogen. The gas is purified by passing through Fe_2O_3 to remove H_2S and then into a mixture of Fe_2O_3. Na_2CO_3 to remove organic sulphur compounds. The purified gas is compressed to 5 to 25 atm and then led through a converter, containing a catalyst.

Catalyst: It consists a mixture of 100 parts of Cobalt, 5 parts of thoria, 8 parts of magnesia, and 200 parts of keiselguhar earth. It maintained at about 200- 300^oC.



Then a mixture of saturated and unsaturated hydrocarbons results as follows;

$$n\,CO + 2n\,H_2 \text{--------}\rightarrow C_nH_{2n} + n\,H_2O$$

$$n\,CO + (2n+1)\,H_2 \text{--------}\rightarrow C_nH_{2n+2} + n\,H_2O$$

The reaction is exothermic, so out coming hot gaseous mixture is led to a cooler, where a liquid resembling crude oil is obtained. The crude oil thus obtained is then fractionated to yield (i) gasoline and (ii) heavy oil. The heavy oil is reused for cracking to get more gasoline.

Bergius process:

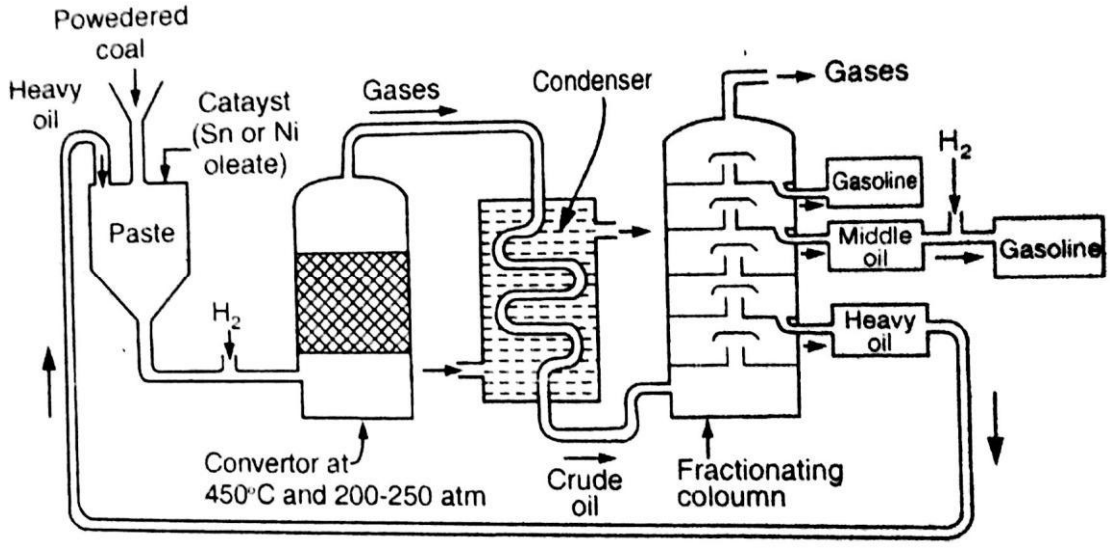
1. The low ash coal is finely powdered and made into a paste with heavy oil and then
   a catalyst composed of tin or nickel oleate is incorporated

2. The whole is heated with hydrogen at 450$_o$C and under a pressure 200-250 atm for
   about 1.5 hours, during which hydrogen combines with coal to form saturated hydrocarbons, which decompose at prevailing high temperature and pressure to yield low boiling liquid hydrocarbons.

3. The issuing gases are lead to condenser, where a liquid resembling crude oil is obtained, which is then fractionated to get (i) gasoline (ii) middle oil (iii) heavy oil.

4. The heavy oil is used again for making paste with fresh coal dust.

5. The middle oil is hydrogenated in vapour-phase in presence of a solid catalyst to
   yields more gasoline.



Gaseous fuels:

Gaseous fuels occur in nature, besides being manufactured from solid and liquid fuels.

The advantages and disadvantages of gaseous fuels are given below :

Advantages:

Gaseous fuels due to erase and flexibility of their applications, possess the following advantages over solid or liquid fuels :

(a) They can be conveyed easily through pipelines to the actual place of need, thereby eliminating manual labour in transportation.

(b) They can be lighted at ease.

(c) They have high heat contents and hence help us in having higher temperatures.

(d) They can be pre-heated by the heat of hot waste gases, thereby affecting economy in heat.

(e) Their combustion can readily by controlled for change in demand like oxidizing or reducing atmosphere, length flame, temperature, etc.

(f) They are clean in use.

(g) They do not require any special burner.

(h) They burn without any shoot, or smoke and ashes.

(i) They are free from impurities found in solid and liquid fuels.

Disadvantages:

(a) Very large storage tanks are needed.

(b) They are highly inflammable, so chances of fire hazards in their use are high.

Some important gaseous fuels:

Here we are going to discuss about the following gaseous fuels.

1. Natural gas 2.Water gas 3. Producer gas 4. Bio gas

5.Liquid Petroleum Gas(L.P.G.)

1. Natural gas:

It is obtained from the wells dug in the earth during mining of petroleum. It is mainly composed of methane and small quantities of ethane along with other hydrocarbons.

If the lower hydrocarbons are present, the gas is called dry gas or lean gas, but if the hydrocarbons having the higher molecular weights are present the gas is known as rich gas or wet gas.

The calorific value of this gas varies from 12000 to 14,000 kcal/m^3

Composition:

Methane – 88.5% Ethane – 5.5%
Propane – 3.7% Butane – 1.8%
Pentane, hydrogen, CO and higher hydrocarbons – 0.5%

Uses:

1. It is an excellent domestic fuel and can be conveyed over very large distances in pipelines.
2. It has recently been used in the manufacture of a number of chemicals by synthetic processes like carbon black, ammonia, methanol and formaldehyde.
3. Methane on microbiological fermentation gives synthetic proteins which are used as animal feed.
4. It is also used for generation of electricity in fuel cells as a source of hydrogen.

Producer gas: ($CO + N_2 + H_2$)

It is a mixture of combustible gases, carbon monoxide and hydrogen associated with large percentageof non-combustible gases, N_2, CO_2 etc.

Composition:

CO – 22 to 30% N_2 – 52 to 55% H_2 – 8 to 12% CO_2 – 3%

Preparation:

It is prepared by passing air mixed with little steam (about 0.35 kg/kg of coal) over a red hot coal or coke bed maintained at about 1100^oC in a special reactor called 'gas producer'. It consists of a steel vessel about 3 m in diameter and 4 m in height. The vessel is lined inside with fire bricks. It is provided with a cup and cone feeder at the top and a side opening for the exit of producer gas. At the base it has an inlet for passing air and steam. The producer at the base is also provided with an exit for the ash formed.

LPG (Liquified Petroleum Gas)

Nowadays LPG has been a common fuel for domestic work and also in most of the industries. The main components of LPG or cooking gas are n-butane, isobutene, butylene and propane (traces of propene and ethane).The hydrocarbon are in gaseous state at room temperature and at atmospheric pressure but can be liquified under higher pressures. The gas can be compressed under pressure in containers and sold under trade names like Indane, Bharat, Petroleum gas, HP gas, etc.

LPG is kept in metallic cylinder attached with burner through pipe. It has two stoppers, one at the cylinder and other at burner. LPG has special odour due to the presence of organic sulphides which are added specially for safety measure. The gas is obtained from natural gas or as a by-product in refineries during cracking of heavy petroleum products.

Characteristics of LPG:

1. It has high calorific value: 27800 kcal/m_3, it is colourless.
2. It gives less CO and least unburnt hydrocarbons. So it causes least pollution.
3. It gives moderate heat which is very good for cooking.
4. Its storage is simple. It is colourless.
5. It has the tendency to mix with air easily.
6. Its burning gives no toxic gases though it is highly toxic.
7. It neither gives smoke nor ash content.
8. It is cheaper than gasoline. It burns with little air pollution and leaves no solid residue. Hence, it is used as fuel in auto vehicles also.
9. It is dangerous when leakage is there. It is highly knock resistant.
10. LPG can be extracted from natural gases and also from refining of crude oil. Cryogenic process is best for the extraction for natural gas.

Advantages of LPG:

1. LPG is used as domestic fuel and as a fuel for internal combustion engines.
2. It is used as a feedstock for the manufacture of various chemicals and olefins by pyrolysis.
3. LPG is used in industries as portable blow lamps, welding, annealing, hardening, steelcuttings, etc.

Knocking:

In an internal combustion engine, a mixture of gasoline vapour and air is used as a fuel. After the initiation of the combustion reaction, by spark in the cylinder, the flame should spread rapidly and smoothly through the gaseous mixture, thereby the expanding gas drives the piston down the cylinder.The ratio of the gaseous volume in the cylinder at the end of the suction-stoke to the volume at the end of compression-stoke of the piston, is known as the 'compression ratio'.

The efficiency of an internal combustion engine increases with the increase in compression ratio, which is dependent on the nature of the constituents present in the gasoline used. In certain circumstances (due to the presence of some constituents in the gasoline used), the rate of oxidation becomes so great that the last portion of the fuel-air mixture gets ignited instantaneously, producing an explosive violence, known as "knocking". The knocking results in loss of efficiently.

Chemical structure and knocking:

The tendency of fuel constituents to knock is in the following order.

Straight-chain paraffins > Branched-chain paraffins > olefins > Cyclo-paraffins > Aromatics

Thus olefins of the same carbon chain length possess better anti-knock properties than the corresponding paraffins and so on.

The knocking is maximum in open chain straight paraffins and least in aromatics. Therefore a fuel to be called a good fuel for the internal combustion engine should have least knocking.

Antiknocking agents

These are the substances added to petrol in order to prevent knocking in I.C. Engines.

Ex: TEL Tetra Ethyl led.
TML Tetra Methyl led.
MTBE Methyl Tertiary Butyl Ether.

Leaded Petrol:

The petrol containing TEL or TML as anti knocking agents is called leaded petrol. TEL or TML are the very good anti knocking agents but has some disadvantages as follows.

a) After combustion lead is deposited as lead oxide on piston and engine walls it leads to mechanical damage.
b) Lead is a poisonous air pollutant.
c) It spoils the catalyst used in catalytic converter.

Unleaded Petrol:

The petrol, which contains antiknocking agent other than lead, is known as unleaded petrol.

Ex: MTBE is used, as an antiknocking agent in place of TEL or TML and the petrol is known as unleaded petrol.

octane number.

Octane number is defined as the percentage of isooctane present in a standard mixture of isooctane and n-heptane, which knocks at the same compression ratio as the petrol being tested.

Isooctane is the branched chain hydrocarbon has least knocking rate, hence its octane number is arbitrarily fixed as 100. N-heptane a straight chain hydrocarbon has highest tendency to knock hence its octane number is fixed as zero. Octane number of petrol is 80 means it contains 80% by volume isooctane and 20% by volume n- heptane.

Cetane number.

It is defined as the percentage of cetane present in standard mixture of a cetane and Alfa- methylnaphthalene, which knocks at the same compression ratio as the diesel fuel being tested.

Power alcohol

A mixture of ethyl alcohol and gasoline blend, which can be used as fuel in internal combustion engine, is known as power alcohol or gasohol.

Absolute alcohol is mixed with ether, benzene etc compounds and one volume of this is mixed with four volumes of petrol and is used as a fuel.

Advantages:

1. The power out put is good.
2. It has better antiknock property.
3. Ethanol is biodegradable; hence it is environmental friendly fuel.
4. The use of ethanol in alcohol increases the oxygen content of the fuels and promotes more and complete combustion of hydrocarbons in gasoline.
5. It reduces carbon monoxide emission.

Calorific value

Calorific value is defined as the amount of heat liberated when a unit mass of fuel is burnt completely in presence of air or oxygen.

Calorific value is of two types as follows:-

1) Higher calorific value. (HCV) or Gross calorific value. (GCV)
2) Lower calorific value. (LCV) or Net calorific value. (NCV)

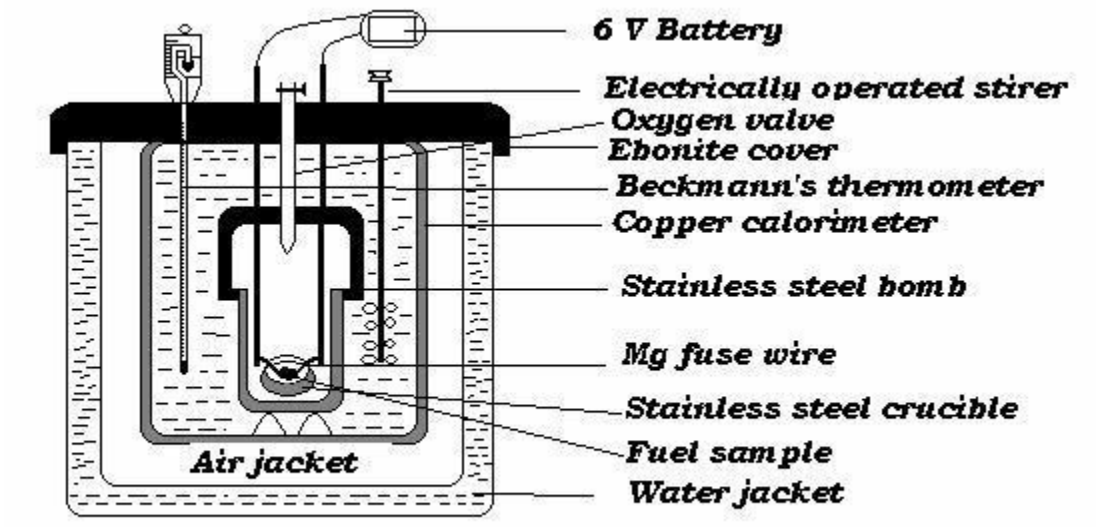
1) HCV: - It is the amount of heat liberated when a unit mass of fuels burnt completely in the presence of air or oxygen and the products of combustion are cooled to room temperature. Here it includes the heat liberated during combustion and the latent heat of steam. Hence its value is always higher than lower calorific value.

2) LCV: - It is amount of heat liberated when a unit mass of fuel is burnt completely in the presence of air or oxygen and the product of combustion are let off completely into air. It does not include the latent heat of steam. Therefore it is always lesser than HCV.

NCV = HCV – Latent heat of steam.

= HCV –0.09X % H_2 X 587 cal/g

Determination of calorific value of solid fuel using Bomb Calorimetric method



Bomb Calorimeter

A small quantity of a fuel is weighed accurately (M Kg) and is placed in the Bomb. The bomb is placed in known amount water taken in a copper calorimeter. The initial temp of water is noted as a t_1^0C with the help of thermometer. Oxygen gas is pumped under pressure 20 to 25 atm through the O_2 valve provided.

The fuel is ignited by passing electric current through the wires provided. As the fuel undergoes combustion and liberates heat, which is absorbed by surrounding water. The water is stirred continuously to distribute the heat uniformly and the final temp attained by water is noted t_2^0C. & gross calorific value of the fuel is calculated as follows

Calculation:

Mass of the fuel = M Kg.
Initial temp of the water = t_1^0C
Final temp of the water = t_2^0C
Change in temp = $\triangle t = (t_2 - t_1)^0 C$
Specific heat of water = S
Water equivalent of calorimeter = W Kg.

$$GCV = \frac{W \times S \times \triangle t}{M} \quad J/Kg \qquad \text{or}$$

$$GCV = \frac{(W+w) \times S \times \triangle t}{M} \quad J/Kg$$

$$NCV = GCV - 0.09 \times \%H_2 \times 587 \ cal/g$$

PROBLEMS:

1) Calculate calorific value coal samples from the following data.

Mass of the coal = 1g.
Water equivalent of calorimeter = 2 Kg.
Specific heat of water = 4.187 J/Kg/c.
Rise of temperature = 4.8^0C.

SOLUTION: $GCV = W \times \frac{S \times \triangle t}{M} \ J/Kg$

$$= \frac{2 \times 4.187 \times 4.8}{0.001}$$

$$= \mathbf{40195.2\ KJ/Kg}.$$

2) A coal sample with 93% carbon, 5% of Hydrogen and 2% Ash is subjected to combustion in a bomb calorimeter. Calculate GCV and NCV Given that. **6 Marks** Mass of the coal sample = 0.95g

Mass of water in copper calorimeter = 2000g.
Water equivalent wt of calorimeter = 700g.

Rise in temp = 2.8^0C
Latent heat of = 587 cal/g. Specific heat
of water = 1 cal/g/^{0}C

$$GCV = \frac{(W+w) \times S \times \triangle t}{M}$$

$$= \frac{(2000+700) \times 10^{-3} \text{ kg} \times 1 \text{ cal/g/}^0\text{C} \times 2.8^0\text{C} \times 4.184}{0.95 \times 10^{-3} \text{ kg}}$$

$$= 33295.83 \text{ J/kg.}$$

$$NCV = GCV - 0.09 \times \%H_2 \times 587 \times 4.184 \text{ J/kg.}$$
$$= 33295.83 \text{ J/kg} - 0.09 \times 5 \times 587 \times 4.184 \text{ J/kg.}$$
$$= \mathbf{32190.62 \text{ J/kg}}$$

3) When 0.84g of coal was burnt completely in Bomb calorimeter the increase in temp of 2655 grams of water was 1.85^0C if the water equivalent calorimeter is 156g Calculate GCV.

$$GCV = \frac{(W+w) \times S \times \triangle t}{M}$$

$$= \frac{(2655+156) \times 1.85 \times 10\text{-}3 \times 4.187}{0.84 \times 10\text{-}3}$$

$$= 25921.26 \text{ J/Kg}$$

4) Calculate GCV and NCV of a fuel from the following data. Mass of fuel=0.75g, W=350g $\triangle$ t =3.02^0C, Mass of water = 1150, % H_2=2.8.

$$GCV = \frac{(W+w) \times \triangle t \times S}{M}$$

$$= \frac{(1150+350) \times 10^{-3} \times 3.02 \times 4.184}{0.75 \times 10^{-3}}$$

GCV = **25271. 36 KJ/Kg**

$$NCV = GCV - 0.09 \times H \times 587 \times 4.184$$
$$= 25271.36 - 0.09 \times 2.8 \times 587 \times 4.184$$

NCV = **24652.44 KJ/Kg**

5) Calculate calorific value of a fuel sample of a coal form the following data. Mass of the coal is 0.6g. Water equivalent wt of calorimeter is 2200g. Specific value 4.187 Kg/KJ/C rise in temperature = 6.52^0C.

$$GCV = \frac{(W_1+W_2) \times S \times \triangle t}{M}$$

$$= \frac{(2200) \times 10\text{-}3 \times 4.184 \times 6.52}{0.6 \times 10\text{-}3}$$

$$= \mathbf{100025.49\ KJ/Kg}.$$

6) Calculate GCV and NCV of a fuel from the following data.

Mass of fuel =0.83g, W=3500g. , W = 385 g, t1 =29.2^{0}C, t_2 = 26.5^{0}C, % H_2 = 0.7 and S = 4.2 kj/kg/c

$$GCV = \frac{(W+w) \times \triangle t \times S}{M}$$

$$= \frac{(3.5 + 0.385) \times (29.2 - 26.5) \times 4.2}{0.83 \times 10^{-3}}$$

GCV = 53079.39 KJ/Kg

NCV = GCV –0.09 x H x 587 x 4.184

= 53079.39 – 0.09 x 0.7 x 587 x 4.2

NCV = **52924.07 KJ/Kg**